Project 97: Smart Campus Incident and Maintenance Prioritization

Python Mini Project - College Assignment

This project analyses hypothetical campus maintenance reports by severity, location importance, and reported frequency. It calculates priority scores, ranks incidents, and produces a summary.

1. Aim

To classify campus maintenance incidents, calculate priority scores, rank complaints, and generate a priority summary using basic Python.

2. Algorithm

1. Start the program.

2. Store incident names, locations, severity, frequency, and location importance.

3. Calculate priority score = severity x 3 + frequency x 2 + location score.

4. Classify incidents: High (18 or above), Medium (12-17), or Low (below 12).

5. Rank incidents in descending order of score.

6. Count incidents in each priority category.

7. Display incident details, ranking, and summary; stop.

3. Priority Rules

Priority Score = (Severity x 3) + (Frequency x 2) + Location Score. High: 18 or above; Medium: 12-17; Low: below 12. These are example assignment rules.

4. Python Program

```
# Project 97: Smart Campus Maintenance Prioritization
print("=====================================")
print("  SMART CAMPUS MAINTENANCE SYSTEM")
print("=====================================")
# Incident details
incidents = [
    "Electrical Problem",
    "Water Leakage",
    "Broken Bench",
    "Internet Issue"
]
# Locations
locations = [
    "Laboratory",
    "Hostel",
    "Classroom",
    "Library"
]
# Severity: 1 = Low, 2 = Medium, 3 = High
severity = [3, 2, 1, 2]
# Number of times reported
frequency = [5, 4, 2, 3]
# Location importance scores
location_score = [3, 2, 2, 3]
# Store priority scores
scores = []
```

```
priority = []
print("\nINCIDENT DETAILS")
for i in range(len(incidents)):
    # Calculate priority score
    score = (severity[i] * 3) + \
            (frequency[i] * 2) + location_score[i]
    scores.append(score)
    # Classify priority
    if score >= 18:
        level = "HIGH"
    elif score >= 12:
        level = "MEDIUM"
    else:
        level = "LOW"
    priority.append(level)
    print("\nIncident:", incidents[i])
    print("Location:", locations[i])
    print("Severity:", severity[i])
    print("Frequency:", frequency[i])
    print("Priority Score:", score)
    print("Priority Level:", level)
# Ranking incidents
ranking = []
for i in range(len(incidents)):
    ranking.append([scores[i], incidents[i]])
ranking.sort(reverse=True)
print("\n=====================================")
print("       INCIDENT PRIORITY RANKING")
print("=====================================")
for i in range(len(ranking)):
    print(i + 1, ".", ranking[i][1],
          "- Score:", ranking[i][0])
# Priority summary
high = 0
medium = 0
low = 0
for level in priority:
    if level == "HIGH":
        high += 1
    elif level == "MEDIUM":
        medium += 1
    else:
        low += 1
print("\n=====================================")
print("          PRIORITY SUMMARY")
print("=====================================")
print("Total Incidents:", len(incidents))
print("High Priority:", high)
print("Medium Priority:", medium)
print("Low Priority:", low)
print("\n=====================================")
print("        REPORT COMPLETED")
print("=====================================")
```

5. Sample Output

```
=====================================
  SMART CAMPUS MAINTENANCE SYSTEM
=====================================
INCIDENT DETAILS
Incident: Electrical Problem
Location: Laboratory
Severity: 3
Frequency: 5
Priority Score: 22
Priority Level: HIGH
Incident: Water Leakage
Location: Hostel
Severity: 2
```

```
Frequency: 4
Priority Score: 16
Priority Level: MEDIUM
Incident: Broken Bench
Location: Classroom
Severity: 1
Frequency: 2
Priority Score: 9
Priority Level: LOW
Incident: Internet Issue
Location: Library
Severity: 2
Frequency: 3
Priority Score: 15
Priority Level: MEDIUM
====================================
       INCIDENT PRIORITY RANKING
====================================
1 . Electrical Problem - Score: 22
2 . Water Leakage - Score: 16
3 . Internet Issue - Score: 15
4 . Broken Bench - Score: 9
====================================
          PRIORITY SUMMARY
====================================
Total Incidents: 4
High Priority: 1
Medium Priority: 2
Low Priority: 1
====================================
        REPORT COMPLETED
====================================
```

6. Flowchart (Text Representation)

```
START
  |
Store incident data
  |
Calculate priority score
  |
Score >= 18?  Yes -> HIGH
  No
Score >= 12?  Yes -> MEDIUM
  No -> LOW
  |
Rank incidents and count priorities
  |
Display report
  |
STOP
```

7. Python Concepts Used

Variables, integers, strings, arithmetic and comparison operators, conditional statements (if/elif/else), for loops, lists, nested lists, and list sorting.

8. Result

The program reports 4 incidents: 1 High, 2 Medium, and 1 Low. The Electrical Problem in the Laboratory ranks first with a score of 22. The rules are illustrative and do not replace real emergency safety procedures.